

ICE BRENT CRUDE OIL FUTURES

Spread & Curve Structure — Fundamental Reference Supply, Demand, Storage, Geopolitics & the Forward Curve

This document presents a comprehensive, fact-based reference on the fundamental drivers of ICE Brent Crude Oil futures (ticker: B), with particular focus on the forces that shape the forward curve — from prompt spreads to the long end. Brent is the world's most important crude oil benchmark, used to price approximately 70-75% of internationally traded crude. The curve structure reflects the balance of physical supply and demand in real time, filtered through storage economics, geopolitical risk, OPEC+ policy, refinery demand cycles, and speculative positioning. Understanding each of these layers is the foundation of any systematic approach to spread and butterfly trading on the Brent curve.

Futures First — Internal Education Material
For Analyst Group Use

1. Contract Specifications and Curve Structure

ICE Brent Crude trades on ICE Futures Europe. Key specifications:

Parameter	Detail
Exchange	ICE Futures Europe (London)
Ticker	B (front month) / BZ on CME Globex
Contract size	1,000 barrels
Quotation	USD per barrel, minimum tick \$0.01 (= \$10 per contract)
Contract months	Monthly, every calendar month out to approximately 8-10 years forward
Settlement type	Cash-settled against the ICE Brent Index (not physical delivery)
ICE Brent Index	Average of 25-day BFOE (Dated Brent) transactions in the 15-day forward window
Trading hours	01:00 – 23:00 London time (Sunday–Friday); liquid hours 07:00–17:00
Expiry	Last trading day = last business day of the month 2 months before delivery month
Open interest	Largest concentration in months 1–6; meaningful liquidity to ~24 months; tradeable to ~60+ months

The cash-settlement mechanism against the Dated Brent index (rather than physical delivery) means the futures price must converge to the physical spot market. This makes the relationship between the futures curve and the physical BFOE market fundamental to understanding spread behaviour near expiry.

1.1 BFOE and the Physical Market

The Brent benchmark is actually a composite of four crude streams collectively known as BFOE:

- **Brent Blend** — North Sea, Sullom Voe terminal, Scotland.
- **Forties** — largest North Sea stream. Higher sulphur (sour). Typically the price-setting stream due to volume.
- **Oseberg** — lighter, sweet. Norwegian North Sea.
- **Ekofisk** — lighter, sweet. Norwegian/UK North Sea.
- **Troll** — added to the basket in 2018 to address declining legacy field production.

The physical BFOE market trades in "15-day" cargoes (seller gives 15 days notice of loading date). The ICE Brent Index averages these transactions in the window 10-25 days forward. Significant divergence between dated Brent and the futures front month creates convergence pressure and directly drives the cash-futures basis, which is the primary driver of prompt spread behaviour.

1.2 Curve Terminology

- **Backwardation** — front month trades at a premium to deferred months. Indicates current physical tightness: immediate demand exceeds available supply. Incentivises stock drawdown, discourages storage. Historically the more common state for Brent.
- **Contango** — front month trades at a discount to deferred. Indicates current physical surplus: supply exceeds immediate demand. Incentivises storage: a trader can buy physical, store it, and sell the forward at a profit (storage play) when the contango exceeds total storage costs (roughly

\$0.30–0.60/bbl/month for tanker storage, \$0.15–0.30/bbl/month for onshore).

- **Flat price** — the outright level of a specific contract month (e.g. Brent Dec \$80/bbl).
- **Time spread** — the price difference between two contract months (e.g. Dec minus Jan). The spread is the core instrument for expressing a view on the supply/demand balance.
- **Fly (butterfly)** — the spread of spreads. Long the C1–C2 spread, short the C2–C3 spread (or equivalently: long C1, short 2×C2, long C3). Expresses a view on the curvature or kink at a specific point on the curve.

2. Supply Fundamentals

2.1 OPEC+ Production Policy

OPEC+ is the most powerful single actor in global crude supply, comprising the 13 OPEC members plus 10 allied non-OPEC producers including Russia, Kazakhstan, and others. Together they produce approximately 40% of global oil output and hold over 80% of proven reserves.

- **Quota system** — OPEC+ sets production quotas for member countries, typically reviewed and adjusted at ministerial meetings. Since 2016, the group has operated under a coordinated framework (the "Declaration of Cooperation"). Quota changes are the single most powerful driver of medium-term Brent curve structure.
- **Voluntary additional cuts** — since 2023, Saudi Arabia and Russia have made unilateral voluntary cuts above and beyond the collective quota. These announcements are market-moving events that steepen backwardation or flatten contango immediately across the prompt curve.
- **Compliance monitoring** — actual compliance with quotas varies significantly. Iraq, Kazakhstan, UAE, and Russia have historically produced above their allocations. Secondary source production estimates (from Kpler, Tanker Trackers, OPEC's own MOMR) are essential to assess actual production vs stated policy.
- **Exemptions** — Libya and Iran have historically been exempt from quotas due to sanctions/conflict. Unexpected resumption of Libyan or Iranian supply at scale is a material bearish event for front spreads.
- **Ministerial meeting outcomes** — OPEC+ holds formal meetings 2–4 times per year, with ad hoc communications between. The market prices in expectations ahead of meetings; the spread between actual outcome and consensus moves the curve substantially. Front spreads are most sensitive; back spreads move less unless the policy change is signalled as structural.

OPEC+ policy changes affect the back curve with a lag: a production cut announced today typically takes 1-2 months to fully manifest in lower supply flows and inventory draws, meaning C3-C6 spreads often price the effect before the prompt month does.

2.2 Non-OPEC Supply

- **United States (tight oil / shale)** — the most important non-OPEC supply variable. US crude production reached approximately 13+ million b/d in 2023–24, making the US the world's largest producer. Shale production responds relatively quickly to price signals (6–9 month lag from drilling decision to first production), making the US rig count a leading supply indicator. WTI Midland — the primary US export grade — is now part of the Dated Brent assessment, directly linking US shale dynamics to the Brent benchmark.
- **Brazil and Guyana** — fast-growing deepwater producers. Brazilian pre-salt output has grown substantially; Guyana (Stabroek block, ExxonMobil) is ramping rapidly and represents the largest new oil province of the 2020s. Both produce medium-sour grades competitive with Brent streams. Long-cycle projects (3–5 year development horizon) — back-of-curve implications are more significant than prompt.
- **Norway** — a major North Sea producer, directly contributing to the BFOE basket. Johan Sverdrup field (500,000+ b/d capacity) transformed Norwegian output. Planned maintenance seasons (typically May–June and October) create recurring, predictable supply dips that historically tighten prompt spreads.
- **Canada (oil sands)** — large volumes of heavy sour crude, landlocked without pipeline access to tidewater for much of its history. TMX expansion (Trans Mountain) now provides Pacific coast export capacity. Canadian supply primarily affects WTI more than Brent directly.
- **Russia** — second-largest OPEC+ producer. Post-2022 sanctions have redirected Russian Urals and ESPO barrels to India and China. Russian export volumes tracked via tanker data; loading delays or

weather at Baltic/Black Sea export terminals create observable supply disruptions.

- **Libya** — the most volatile producer globally. Civil conflict, port blockades, and political fragmentation cause frequent, unpredictable production swings of 200,000–800,000 b/d. Libyan supply outages create immediate tightening in front spreads; resumptions of supply create equally sharp reversals.

2.3 Field Decline Rates and the Base Effect

Global conventional oil fields decline naturally at approximately 4–6% per year without new investment. This "base decline" means the industry must invest continuously just to maintain flat production. The back of the Brent curve implicitly prices this treadmill: without sufficient upstream investment, long-dated deferred contracts embed a structural supply-scarcity premium.

- Legacy North Sea fields (Forties, Brent, Ekofisk) have been in decline for decades. Total North Sea output (UK + Norway) peaked in the late 1990s–2000s and has declined substantially, requiring the BFOE basket to be periodically expanded (Troll added 2018) to maintain contract relevance.
- The IEA tracks global upstream capex annually — below-trend investment years (2015–16, 2020) have historically preceded supply shortfalls 3–5 years later, which directly affects back-curve pricing.

2.4 Strategic Petroleum Reserves (SPR)

Governments hold emergency crude reserves that can be released to market during supply disruptions:

- **US SPR** — the world's largest strategic reserve (~700 million barrels capacity, though significantly drawn down after 2021–22 releases). US SPR releases are bearish for prompt Brent and tend to flatten or invert the front spread, since they add supply temporarily without changing the underlying structural balance. Subsequent refill of the SPR is a bullish prompt demand signal.
- **IEA coordinated releases** — the International Energy Agency can coordinate emergency releases from member country reserves. Historical precedents: Gulf War (1991), Libyan conflict (2011), Russia-Ukraine (2022). These are typically one-off events that temporarily suppress the prompt spread.
- SPR release/refill cycles create predictable, medium-term spread implications that are distinct from structural supply/demand changes. The market typically prices them as temporary, so their effect is more pronounced at C1–C3 than C6+.

3. Demand Fundamentals

3.1 Global Demand Structure

Global oil demand runs at approximately 100–104 million barrels per day (2023–24). Understanding where demand comes from and how it varies seasonally and cyclically is essential for curve analysis.

Demand Category	Share of Global Demand	Growth Driver	Curve Sensitivity	Key Signal
Road transport (gasoline/diesel)	~45%	Vehicle fleet size, economic activity	Prompt + seasonal	US weekly product demand, PMI data
Aviation (jet fuel/kerosene)	~8%	Passenger & cargo traffic	Prompt + seasonal	IATA traffic data, airline capacity
Petrochemicals (naphtha)	~15%	Plastics, chemicals production	Mid-curve	Chemical industry PMI, ethylene margins
Industrial & power generation	~12%	Industrial output, gas substitution	All curve	Manufacturing PMI, gas prices
Shipping (bunker fuel)	~5%	Global trade volumes	Prompt	Baltic Dry Index, container shipping rates
Heating oil / residential	~7%	Temperature, gas/oil switching	Seasonal/prompt	Weather forecasts, HDD/CDD data
Other (military, agriculture)	~8%	Misc.	Stable	—

3.2 China — The Dominant Demand Variable

China accounts for approximately 14–16% of global oil consumption and has been responsible for the majority of global demand growth since 2000. Chinese demand data is the single most important demand-side variable for the Brent forward curve, particularly the mid-curve (C4–C12).

- **Apparent demand** — derived from crude imports plus domestic production minus changes in inventories. Monthly data from China's customs agency (GACC) and National Bureau of Statistics (NBS). Published 3–4 weeks after month end.
- **Crude oil imports** — the most timely signal. Monthly GACC data shows total crude imports by volume; tanker tracking services (Kpler, Vortexa) provide near-real-time estimates. Chinese imports running above or below consensus expectations directly moves mid-curve spreads.
- **Teapot refineries** — independent Chinese refineries (primarily in Shandong province) process significant volumes, often importing sanctioned barrels (Russian ESPO, Iranian Light). Their import quotas and utilisation rates are a key variable in Chinese crude demand.
- **Strategic petroleum reserve (SPR) building** — China does not publish SPR data officially, but analysts estimate SPR builds/draws from the difference between imports and apparent consumption. SPR build phases are bullish for mid-curve; release phases are bearish.
- **Economic indicators** — China's manufacturing PMI, vehicle sales, industrial production, and property sector health all serve as leading indicators for Chinese oil demand. Weakness in the property sector (post-2021) has been a persistent structural headwind for Chinese diesel demand.

3.3 OECD Demand Seasonality

OECD oil demand follows well-established seasonal patterns that create recurring, predictable pressures on the Brent curve:

- **Q4 (Oct-Dec) — peak demand season:** Northern Hemisphere heating oil demand builds. US gasoline demand remains solid through end of driving season. Aviation demand elevated from holiday travel. Historically the strongest period for prompt Brent and backwardation building.
- **Q1 (Jan-Mar) — seasonal trough:** Post-holiday demand drop. Refinery maintenance season begins (reducing crude runs). Northern Hemisphere winter heating partially offsets. Typically the weakest seasonal demand period; curve often softens.
- **Q2 (Apr-Jun) — demand recovery:** US driving season builds. European summer travel begins. Refinery runs increase as maintenance concludes. Prompt spreads typically firm seasonally.
- **Q3 (Jul-Sep) — summer peak:** Peak gasoline demand (US driving season in full swing). Power generation demand elevated in Middle East/Asia. Aviation demand at annual peak. Backwardation tends to build or sustain.

3.4 Refinery Demand — The Transmission Mechanism

Crude oil demand is ultimately derived demand — crude is purchased to be processed into products. Refinery economics and runs are the transmission mechanism between end-user demand and crude oil pricing:

- **Global refinery utilisation** — when utilisation is high (85%+), crude demand is robust and prompt spreads tend to firm. When refineries take maintenance or cut runs, crude demand softens.
- **Crack spreads** (refinery margin) — the spread between product prices (gasoline, diesel, jet fuel) and crude input cost. Wide crack spreads incentivise refineries to maximise crude runs, supporting prompt demand. Collapsing crack spreads lead to run cuts, weighing on the front of the Brent curve.
- **North Sea planned maintenance** — UK and Norwegian North Sea fields have published maintenance programmes, typically concentrated in May-June and September-October. These create predictable, calendar-visible tightening in the physical BFOE market and directly affect prompt spreads.
- **Asian refinery turnarounds** — Asian (particularly Chinese and Indian) refinery maintenance seasons create lumpy drops in crude import demand, typically affecting the prompt curve within 3-6 weeks.

4. Storage Economics and the Cost of Carry

Unlike agricultural markets, crude oil can be stored in large quantities in pipelines, tanks, and tankers. Storage economics are the primary determinant of the carry structure of the Brent forward curve — especially at the front end (C1–C6).

4.1 The Storage Carry Model

In a full-carry contango market, the price difference between months reflects the total cost of storing a barrel:

Storage Type	Cost (approx.)	Capacity	Key Locations	Spread Implication
Onshore tank storage	\$0.15–0.30/bbl/month	~3.5 bn bbl globally	Cushing, ARA, Fujairah, Singapore, Saldanha Bay	Fundamental carry floor — contango must exceed this to incentivise tank fills
Floating storage (VLCC)	\$0.30–0.60/bbl/month	Flexible, 2–5 mn bbl/vessel	Offshore Singapore, Persian Gulf, North Sea	Used in deep contango; high tanker rates reduce floating storage economics
Pipeline linefill	Minimal variable cost	Fixed, infrastructure-dependent	North Sea pipelines	Not arbitrageable by traders; part of structural inventory
Strategic reserves	Government-held	~4.1 bn bbl IEA nations	US Gulf Coast, Europe, Japan, Korea	Release creates temporary bearish shock to prompt; not economically motivated

- **Full carry contango:** when the forward curve rises steeply enough to cover all storage costs plus financing, tanks fill and floating storage activates. This was seen acutely in April 2020 (COVID demand collapse), when onshore storage was near capacity and the WTI front month briefly went negative.
- **Sub-carry contango:** the curve is in mild contango but below full carry cost. No additional storage incentive; existing holders draw down tanks at a measured pace.
- **Backwardation:** the curve discourages storage. Holding physical oil has a negative roll cost — each passing month the holder is selling at a premium vs where deferred was priced. This incentivises inventory drawdowns and just-in-time physical procurement.

4.2 Inventory Data — The Weekly Heartbeat

Crude oil inventory data is published more frequently than for any other commodity, and the weekly cycle is the primary data rhythm for front-spread traders:

Report	Publisher	Frequency	Release (US time)	Key Data Points	Curve Impact
EIA Weekly Petroleum Status	US EIA	Weekly	Wed 10:30 ET	US crude stocks, Cushing stocks, product stocks, refinery runs, imports/exports	Immediate move in prompt Brent; largest single weekly data event
API Weekly Statistics	API	Weekly	Tue ~16:30 ET	Crude, gasoline, distillate stocks — private sector estimate, released night before EIA	Sets expectations; surprise vs EIA creates intraday moves

EIA Monthly STEO	US EIA	Monthly	2nd week	Global supply/demand balance forecast, OECD inventory projections	C3-C12 re-pricing on major revisions
IEA Oil Market Report (OMR)	IEA	Monthly	~Mid-month	Global demand forecast, supply estimates, OECD stock levels, refinery data	Medium-term curve — significant if IEA revises demand meaningfully
OPEC Monthly Oil Market Report	OPEC Sec.	Monthly	~2nd week	Member production data (secondary sources), demand forecast, supply outlook	OPEC's own view of the balance — moves C3-C12
JODI Oil World Database	JODI	Monthly	~25th of month	60+ country inventory and production data	Comprehensive but lagged; used for structural confirmation
Kpler / Vortexa / Refinitiv	Private	Daily/Weekly	Continuous	Real-time tanker tracking, loading data, port congestion, crude flows by grade	Most timely signal — used by physical traders and informed desks

The EIA Wednesday report is the single most market-moving scheduled data event for front Brent spreads. A surprise crude stock build of 5+ million barrels can move the C1-C2 spread by \$0.10-0.30/bbl within minutes. Cushing stocks (the WTI delivery point) are also watched even by Brent traders due to the WTI Midland inclusion in BFOE.

4.3 OECD Commercial Inventories — The Structural Indicator

The IEA tracks OECD commercial crude and product inventories, reported in both absolute terms (million barrels) and relative to a five-year seasonal average. This metric is the most widely used structural indicator for medium-term curve positioning:

- OECD stocks significantly below the 5-year average (demand days of cover): structural backwardation signal, particularly for C3-C12.
- OECD stocks at or above 5-year average: contango pressure. OPEC+ often references this metric explicitly in production decisions.
- The transition from above-average to below-average stocks (or vice versa) is a regime shift that reprices the entire curve — not just the prompt. These transitions are the best medium-term spread entries.
- Floating storage volumes (tracked by tanker analytics firms) supplement onshore inventory data, particularly during periods of supply surplus when onshore tanks are full and excess crude sits offshore.

5. Geopolitical Risk and Supply Disruption

Geopolitical risk is a permanent feature of crude oil markets. It affects the Brent curve in a specific way: geopolitical risk premia are concentrated in the front months (C1–C3) and decay rapidly along the curve because the market prices resolution risk — if a disruption persists, it shifts from a risk premium to a structural supply change. If it resolves, the premium evaporates instantly.

5.1 The Strait of Hormuz

The single most important geopolitical chokepoint for global crude flows. Approximately 20–21 million barrels per day pass through the Strait, representing roughly 20–25% of global oil consumption and including the vast majority of exports from Saudi Arabia, UAE, Kuwait, Iraq, and Qatar. There is no short-term alternative route for most of this volume (the Saudi East-West pipeline capacity is ~5 million b/d).

- Any credible military threat to the Strait (Iran-US tensions, shipping attacks) immediately adds a significant risk premium to front-month Brent and steepens prompt backwardation.
- The market's sensitivity depends on context: during periods of high spare capacity (OPEC+ sitting on 5+ mb/d of cuts), the Hormuz premium is smaller because alternative supply could theoretically replace flows. During periods of low spare capacity, the premium is larger.
- Tanker attacks in the Gulf of Oman and Red Sea (2019 Abqaiq attack, 2023–24 Houthi attacks on Red Sea shipping) create secondary effects: higher insurance costs, route diversions, and shipping delays that tighten the physical crude market independently of production volumes.

5.2 MENA Production Risk — Key Country Profiles

Country	Production (approx.)	Export Volume	Risk Profile	Spread Impact
Saudi Arabia	9–12 mb/d	6–7 mb/d	Low operational risk; high policy risk (OPEC+ decisions, Vision 2030)	Announcement of voluntary cuts = immediate prompt tightening
Iraq	4.2–4.5 mb/d	3.5 mb/d	Medium — export infrastructure (Basra) vulnerable; Kurdish region disputes	Kirkuk pipeline outages historically tighten C1–C2 briefly
UAE	3.0–3.5 mb/d	2.5 mb/d	Low — highly stable, Murban grade added to BFOE pricing basket	Steady; UAE pushing for higher quota allowances within OPEC+
Kuwait	2.5–2.7 mb/d	2.0 mb/d	Low — stable, all exports through Hormuz	Minimal independent signal; moves with Saudi Arabia
Iran	3.0–3.5 mb/d	1.5–2.0 mb/d	High sanctions risk; nuclear negotiations determine export ceiling	Sanction enforcement tightening or relaxation = structural curve shift
Libya	1.0–1.3 mb/d	0.9–1.1 mb/d	Very high — civil conflict, political fragmentation, NOC disputes	Frequent outages 200–800 kb/d; immediate prompt impact; fast reversal
Russia	9.5–10 mb/d	7–8 mb/d	High sanctions risk; export redirected post-2022; western price cap enforcement	Shadow fleet disruptions / insurance issues create occasional tightness

5.3 The Red Sea / Suez Canal Route

The Suez Canal handles approximately 10% of global trade by value, including significant crude and product flows from the Middle East to Europe. Disruptions to the Red Sea route (Houthi attacks,

2023–24) force rerouting around the Cape of Good Hope, adding 10–14 days to transit times.

- Longer voyage times increase floating storage (oil in transit), temporarily reducing the effective available supply that reaches European refineries, which tightens the North Sea / European physical market and supports prompt Brent spreads.
- Higher shipping costs (insurance, fuel, vessel time) raise the landed cost of Middle Eastern crude in Europe, widening the effective Brent premium and supporting the North Sea BFOE physical basis.

5.4 Iranian Supply — The Sanctions Variable

Iran holds the world's fourth-largest proven crude reserves and was producing approximately 3.8 mb/d before the 2018 sanctions reimposition. The trajectory of US-Iran relations and nuclear negotiations creates one of the most binary supply variables in the market:

- Nuclear deal restoration (Joint Comprehensive Plan of Action / JCPOA): would theoretically allow Iran to increase exports by 1–1.5 mb/d within 6–12 months. Progress toward a deal is immediately bearish for the back curve, as the market prices additional medium-term supply.
- Tighter sanctions enforcement: reduces Iranian exports, particularly the "dark fleet" barrels flowing to China and India. Bullish for prompt and mid-curve as sanctioned barrels leave the accessible market.
- Iran has maintained significant exports despite sanctions via the "shadow fleet" — vessels that disable AIS transponders, use ship-to-ship transfers, and falsify documentation. The scale of this leakage is tracked by tanker analytics services and affects the balance of the physical market.

6. Macro, Financial, and Cross-Market Drivers

6.1 The US Dollar

Crude oil is priced in USD globally. A stronger USD makes oil more expensive in local currency terms for non-US buyers, which typically reduces demand at the margin, applying bearish pressure to the prompt curve. Conversely, a weaker USD supports crude prices by increasing purchasing power of non-dollar economies.

- The USD index (DXY) has a well-documented negative correlation with Brent flat price. This correlation is strongest for the outright and weaker for spreads (which reflect supply/demand balance, not currency-adjusted demand level).
- For spread traders, USD is less important than for outright traders, but extreme USD moves (particularly emerging market currency crises) can reduce demand in EM economies and shift the forward balance sheet, affecting C3-C12 pricing.

6.2 Interest Rates and the Carry Cost

Rising interest rates increase the financing cost of holding physical crude oil, which has two effects on the forward curve:

- Storage economics worsen: at higher interest rates, the carry cost of holding physical oil increases, reducing the incentive for contango storage plays and compressing the contango structure.
- The full-carry contango level rises with interest rates: the forward curve must be steeper to incentivise storage when financing costs are higher. This was visible in 2022-23 as aggressive Fed rate hikes compressed the contango simultaneously with supply tightening from OPEC+ cuts.

6.3 Speculative Positioning — COT and Managed Money

Speculative positioning in Brent futures is tracked by the ICE Commitments of Traders report, published weekly (Fridays, with a 3-day lag). Key categories:

- **Managed money (hedge funds, CTAs)** — the largest speculative category. When managed money net long positioning reaches extreme levels (top decile historically), it signals crowding and elevated reversal risk. Conversely, extreme net short positioning has historically preceded short-covering rallies. This is a sentiment indicator, not a fundamental signal.
- **Commercial (producer hedging)** — net short positions represent oil producers selling forward production. Heavy producer hedging in deferred months signals producers expect prices to remain elevated and are locking in. This adds known selling pressure to deferred contracts and can suppress back-curve spreads.
- **Swap dealers** — typically representing financial institutions that hold risk-management positions. Their net position reflects OTC hedging demand from producers, consumers, and financial products.
- **Open interest by month** — the distribution of open interest along the curve indicates where market participants are concentrated. A sharp increase in open interest in a specific deferred month often precedes positioning ahead of a known catalyst.

COT data is a lagging indicator — it reflects positions as of Tuesday close, published Friday. For fast-moving markets, it is most useful as a contrary indicator at extremes, not as a real-time directional signal.

6.4 The Brent-WTI Spread

The spread between ICE Brent and NYMEX WTI is one of the most important cross-market relationships in energy:

- **Structural Brent premium:** Brent typically trades at a premium to WTI because it is light-sweet, seaborne, and globally accessible. WTI is a landlocked US benchmark and reflects domestic US supply/demand dynamics.
- **WTI Midland inclusion in BFOE (since 2023):** the addition of US WTI Midland to the Dated Brent basket has narrowed the Brent-WTI spread structurally and created a more direct link between US production and Brent pricing.
- **Widening Brent premium (Brent >> WTI):** typically signals either geopolitical risk in the Middle East (adding risk premium to Brent) or Cushing inventory builds (weighing on WTI specifically). Wide spread incentivises US crude exports to Europe.
- **Narrowing Brent premium / WTI at par with Brent:** signals strong US export demand or European crude surplus. Historically associated with periods of high US production and export capacity.

6.5 Product Crack Spreads and Refinery Economics

- **Gasoline crack (Naphtha/RBOB vs Brent)** — wide gasoline crack incentivises refineries to maximise crude runs, supporting crude demand and prompt spreads.
- **Diesel/gasoil crack (ICE Gasoil vs Brent)** — the gasoil crack is arguably more important than gasoline for Brent because European and Asian demand is more diesel-oriented. A wide gasoil crack supports refinery crude intake.
- **Fuel oil crack (HSFO vs Brent)** — historically negative (fuel oil cheaper than crude). IMO 2020 sulphur regulations (effective Jan 2020) permanently changed the fuel oil market, reducing demand for high-sulphur fuel oil and affecting sour crude differentials.
- **3-2-1 crack spread** — simplified refinery margin indicator: $(2 \times \text{gasoline} + 1 \times \text{heating oil} - 3 \times \text{crude}) / 3$. When this spread is wide, refineries are incentivised to run at maximum capacity, supporting crude demand.

7. Forward Curve Shape — What Drives Each Part

The Brent forward curve can be divided into zones, each dominated by different drivers. Understanding this zoning is fundamental to spread trading.

Zone	Contracts	Primary Driver	Secondary Driver	Typical Volatility	Key Signal
Prompt	C1-C2	Physical BFOE market, certified stocks, Dated Brent basis	Weather, unplanned outages, SPR	Highest	Dated Brent vs futures basis, tanker lineups
Near-term	C2-C6	EIA/IEA inventory data, OPEC+ compliance, refinery runs	Seasonal demand, crack spreads	High	OECD stock vs 5yr avg, EIA weekly
Mid-curve	C6-C12	OPEC+ production policy, Chinese demand outlook	ENSO, capex cycle, SPR refill	Moderate	OPEC MOMR, IEA OMR revisions, China imports
Back curve	C12-C24	Long-run supply investment, energy transition pace	Structural demand growth rate	Lower	Upstream capex data, IEA long-term scenarios
Long-dated	C24+	Energy transition policy, peak oil demand expectations	OPEC+ long-term strategy	Lowest	OPEC+ long-term strategy, IEA WEO scenarios

7.1 Prompt Spread Mechanics

- The C1-C2 spread (the "nearby spread" or "prompt spread") is the most liquid spread on the curve. It reflects the instantaneous balance between physical BFOE availability and refinery demand.
- When dated Brent trades at a premium to the front futures (the "EFS" or Exchange of Futures for Swaps is positive), physical demand is strong and the prompt spread tends to be in backwardation.
- Scheduled North Sea loading programmes (published monthly by operators) allow the market to anticipate physical availability 4–6 weeks ahead, creating a visible supply schedule that anchors prompt spread expectations.
- The 15-day Brent trading window (where physical cargoes are traded) directly feeds into the ICE Brent Index calculation. When BFOE physical cargoes are over-bid or under-bid in this window, the signal is immediately transmitted to front spread pricing.

7.2 The Fly and Curve Curvature

Butterflies on the Brent curve capture the curvature at specific points and are driven by the market's assessment of supply or demand dynamics at that specific time horizon relative to the periods on either side:

- **Positive fly (hump):** the middle month is cheap relative to its neighbours. Typically indicates a specific, time-limited tightness or surplus at that point — e.g. a known refinery maintenance period reducing demand, or a specific OPEC+ cut expiry date creating a supply step-change.
- **Negative fly (bowl):** the middle month is expensive relative to its neighbours. Indicates an expected supply shortfall centred on that specific month, e.g. peak summer demand, North Sea maintenance, or a specific geopolitical event.
- OPEC+ meeting dates and cut extension/rollover decisions are the most important fly drivers. If a cut is known to expire in month C4, the market will price a "kink" in the curve at C4 — backwardation before the expiry, flatter or contango after. A butterfly centred on C4 captures this directly.
- North Sea maintenance schedules create predictable seasonal flies. If a large North Sea field is scheduled for 4 weeks of maintenance in June, the June fly will be negative (June cheap) — a recurring, calendar-visible opportunity.

7.3 The Back Curve and Energy Transition

Long-dated Brent contracts (C24+) are increasingly influenced by expectations about the energy transition and peak oil demand:

- IEA World Energy Outlook scenarios project peak oil demand at different dates depending on policy pathway (current policies vs net zero). These scenarios, updated annually, are the primary reference point for long-dated curve pricing by financial participants.
- OPEC's World Oil Outlook has consistently projected later and higher peak demand than the IEA. The tension between these two views is directly embedded in the slope of the long end of the Brent curve.
- Upstream investment decisions by majors (capex guidance announcements) influence long-dated deferred pricing: if major producers are reducing exploration capex, the market prices a future supply shortfall into the back curve.
- Carbon pricing and energy transition policy (EU ETS carbon price, US IRA incentives) affect the competitive economics of oil-based products at the margin in the long run, though these are second-order effects at the typical spread trading horizon (1–24 months).

8. Crude Quality, Grade Differentials and Basis

Crude oil is not a homogeneous commodity. Different crude grades vary significantly in quality characteristics that affect their value to refineries, and understanding these differentials is essential for interpreting the relationship between ICE Brent and other benchmarks.

8.1 Key Quality Parameters

Parameter	Definition	Market Significance	Brent/BFOE Level
API Gravity	Measure of density (higher = lighter)	Lighter crudes (38–42° API) yield more high-value products (gasoline, jet fuel)	~38–40° API (light)
Sulphur content	% sulphur by weight	Low sulphur ("sweet") crudes command premium due to lower refining complexity	<0.5% (sweet)
TAN (acidity)	Total Acid Number	High TAN requires special refinery units; discounts naphthenic crudes	Low (non-issue for BFOE)
Viscosity	Resistance to flow	Higher viscosity requires heated transport; affects tanker economics	Low-moderate
Pour point	Temperature at which crude stops flowing	Relevant for Arctic/cold-weather production logistics	Low — not problematic

8.2 Key Crude Grade Relationships to Brent

- **WTI Midland vs Brent (EFP — Exchange of Futures for Physical)** — now embedded in the BFOE basket. The WTI Midland differential to Brent reflects US export economics, Gulf Coast refinery demand, and pipeline logistics. Changes in this differential directly affect the ICE Brent Index.
- **Dubai/Oman (sour crude benchmark)** — the primary Middle Eastern crude benchmark for Asian buyers. The Brent-Dubai spread (EFS) measures the premium of sweet crude over sour, which reflects the global balance between sweet and sour supply, and the refinery configuration mix globally.
- **Urals (Russian medium-sour)** — traditionally priced at a small discount to Brent. Post-2022 sanctions, Urals trades at a substantial discount (historically \$20–35/bbl). Urals price dynamics affect the competitive positioning of Middle Eastern and African crudes in the European market.
- **Murban (UAE, ADNOC)** — added to BFOE-related pricing since 2021. Murban trades on the Abu Dhabi Derivatives Exchange (ADX). Its inclusion in Platts/Argus Brent assessments reflects the growing importance of Gulf production in the Brent price formation.
- **Forties differential** — Forties is typically the cheapest BFOE stream (higher sulphur, more sour). Its differential to Brent flat price influences the Dated Brent assessment and the ICE Brent Index, making it relevant for front contract pricing.

8.3 IMO 2020 and Fuel Oil Specification Change

The International Maritime Organization's 2020 sulphur cap (reducing marine fuel sulphur from 3.5% to 0.5%) was the most significant structural change to crude quality economics in decades:

- Dramatically reduced demand for high-sulphur fuel oil (HSFO), widening the discount for sour crude grades relative to sweet.
- Increased demand for very low sulphur fuel oil (VLSFO) and marine gasoil, which compete with the gasoil/distillate pool, indirectly supporting gasoil crack spreads.
- Middle East sour crude grades (Arab Light, Arab Medium) faced structural pricing headwinds initially, though the effect normalised as refineries adapted.

9. Documented Seasonal Patterns in Brent

While Brent is influenced by a large number of non-seasonal variables (geopolitics, OPEC+ decisions, macro shocks), certain seasonal patterns are statistically robust and should be part of every spread trader's baseline:

Period	Seasonal Driver	Typical Curve Effect	Key Variables to Monitor	Reliability
Jan-Mar	Refinery maintenance season begins; post-Q4 demand trough; cold weather heating demand	Prompt spreads soften; curve often flattens or moves toward contango	EIA runs data, heating degree days, OECD stocks vs 5yr avg	Moderate
Apr-May	Refinery maintenance ends; US driving season begins; crude runs ramp up	Prompt backwardation typically builds; C1-C2 spread strengthens	Gasoline crack spreads, EIA product demand, refinery utilisation	Moderate-High
May-Jun	North Sea maintenance season peak; Norwegian/UK field planned shutdowns	BFOE physical tightens; Dated Brent premium; prompt spread supports	Published North Sea loading programmes, BFOE cargo nominations	High — calendar
Jul-Aug	Peak global demand (gasoline US, power gen Asia/Middle East, aviation)	Backwardation typically at seasonal maximum; prompt very firm	US driving data (EIA), Asian crude imports, Middle East domestic burn	Moderate-High
Sep-Oct	Second North Sea maintenance window; US driving season ends; refinery switch	Brief softening of prompt spread as maintenance reduces BFOE supply AND crude runs	North Sea programmes, US product stock transition (gasoline→heating oil)	Moderate
Nov-Dec	Peak heating demand; Q4 inventory build or draw defines year-end balance	Highly variable — depends on OPEC+ stance; historically prompt supports	Heating degree days, OPEC+ meeting outcome, year-end positioning	Low-Moderate (OPEC dependent)

Important caveat: Brent seasonal patterns are the weakest of the major commodity markets because OPEC+ production policy, geopolitical events, and macroeconomic cycles can completely override seasonal tendencies in any given year. Seasonal patterns should be used as a baseline probability framework, not as a primary signal. Always assess current-year supply/demand balance first.

10. Key Data Release Schedule and Sources

Report	Publisher	Frequency	Release Timing	Primary Curve Impact
EIA Weekly Petroleum Status Report	US EIA	Weekly	Wed 10:30 ET	Prompt C1-C3 — largest single weekly event
API Weekly Statistical Bulletin	API	Weekly	Tue ~16:30 ET	C1-C2 — sets expectations for EIA
EIA Short-Term Energy Outlook (STEO)	US EIA	Monthly	2nd Tue of month	C3-C12 supply/demand balance revision
IEA Oil Market Report (OMR)	IEA	Monthly	~2nd week of month	C3-C12 demand forecast, OECD stocks
OPEC Monthly Oil Market Report	OPEC Secretariat	Monthly	~2nd week of month	C3-C12 OPEC view of balance, member production
OPEC+ Ministerial Meeting	OPEC+	2-4x per year	Variable	All months — most market-moving scheduled event
EIA Monthly Drilling Productivity Report	US EIA	Monthly	~2nd Mon of month	C6-C18 — US shale supply outlook
North Sea loading programmes	Field operators	Monthly	~20th of prior month	Prompt C1-C3 — BFOE physical supply visibility
Ivorian / tanker tracking data	Kpler, Vortexa, Refinitiv	Daily/Weekly	Continuous	Physical crude flows — prompt and mid-curve
JODI Oil World Database	JODI / IEF	Monthly	~25th of month	C3-C6 structural confirmation
ICE COT (Commitments of Traders)	ICE / FCA	Weekly	Fri afternoon (3-day lag)	Sentiment indicator — all months
ENSO update (CPC)	NOAA CPC	Monthly	~Mid-month	C6-C18 — global demand / supply weather outlook
IEA World Energy Outlook	IEA	Annual	October	C24+ — energy transition / long-run demand signal
OPEC World Oil Outlook	OPEC	Annual	November	C24+ — OPEC long-run supply/demand scenario

ICE Brent Crude is the world's most complex commodity market — a simultaneously physical, financial, geopolitical, and macro asset. The forward curve integrates all of these dimensions at once: the prompt spread reflects the immediate physical balance; the mid-curve reflects OPEC+ policy credibility and inventory trajectory; the back curve reflects structural supply investment and demand transition expectations. No single factor determines curve shape — the skill lies in correctly weighting these dimensions at each point in time.

For spread traders, the most reliable edge comes from identifying when one of these dimensions is being over- or under-weighted by the market: a geopolitical event that moves only the prompt without a structural supply change, an inventory signal that the back curve has not yet priced, or an OPEC+ announcement whose credibility differs from what the curve implies. The facts in this document are the toolkit. The edge is in their synthesis.